Supplementary Information for

PDR: A paired diagnosis-and-repair framework for 8-bit object detectors under image corruption

Dinh Thuan Nguyen[1], Lam Phuong Nguyen[1], Vinh Huy Nguyen[2], Sy Vu Quang[3], Mohan Rajesh Elara[4], Anh Vu Le[3,*]

[1]Faculty of Electrical and Electronics Engineering, Ton Duc Thang University, Vietnam

[2]Center of Visualization and Simulation, Duy Tan University, Vietnam

[3]Advanced Intelligent Technology Research Group, Ton Duc Thang University, Vietnam

[4]ROAR Lab, Singapore University of Technology and Design, Singapore

[*]Corresponding author: leanhvu@tdtu.edu.vn

S1 Purpose and scope

This supplement documents the estimands, frozen condition schedule, arm registry with terminology, and inference contract (S2–S5); the calibration-baseline schedule, decision rules, and contrasts (S6–S7); the bootstrap and sensitivity analyses (S8–S9); the evidence layers with the historical exploratory analysis (S10); and the artifact boundary (S11). Sections S12–S15 report post-hoc deployment audits, activation statistics, max-calibration counterfactuals, and the retrained KITTI holdout; these are labeled post hoc and are not part of the frozen protocol. S16 gives a reporting checklist. All reported values are generated deterministically from retained ledgers.

S2 Estimands

For dataset d, model m, corruption condition c, and a pair of treatments (A, B) sharing the same evaluator:

$$G_0 = \mathrm{AP}_A(\mathrm{J95}) - \mathrm{AP}_B(\mathrm{J95}), \tag{S1}$$

$$G_{\mathrm{corr}} = \overline{\mathrm{AP}_A(c) - \mathrm{AP}_B(c)} \quad \text{(equal weight over the 12 cells)}, \tag{S2}$$

$$\Delta E = G_{\mathrm{corr}} - G_0. \tag{S3}$$

J95 denotes the codec-matched clean control: the J95 clean control and every corrupted input are terminally encoded as JPEG quality 95 with chroma subsampling disabled, so the clean control and the corrupted inputs differ only in the named corruption transform, not in container encoding; the separate original-clean cell keeps the dataset-native bytes; it checks control fidelity and anchors the clean-accuracy comparisons behind the RetinaNet collapse and recovery claims, while the paired interaction uses the J95 basis. $\Delta E > 0$ means the A-minus-B gap widens under corruption; $\Delta E < 0$ means it narrows. A narrowing gap near a shared accuracy floor is not read as robustness: absolute corrupted AP is reported alongside every interaction (main text, Section 5).

S3 Frozen condition schedule

Each dataset×model block evaluates 14 conditions on identical image ordering across arms (13 for the original RetinaNet FP32, FP8-legacy, and INT8-legacy arms, which predate codec control):

- clean (original bytes) and J95 (JPEG-95 clean control);
- Gaussian noise: $\sigma \in \{8, 20, 40\}$ on the 0–255 RGB scale;
- motion blur: kernel lengths $\{5, 13, 23\}$ pixels;
- synthetic fog: (strength, blur radius in pixels) $\in \{(0.18, 6), (0.38, 12), (0.62, 20)\}$;
- JPEG compression: qualities $\{70, 40, 15\}$.

Corruption parameters and the JPEG-95 terminal encoding are frozen in versioned configuration `corruptions.json`, whose SHA-256 is recorded as `generator_config_sha256` in the corruption-aware and codec-control calibration manifests; the frozen protocol file `nn_paired_protocol_v1_20260924` fixes the condition list and evaluation contract on top of that generator. The same transform code produces both the evaluation caches and the corruption-aware calibration images.

S4 Treatments and arm registry

The RetinaNet and FCOS arms are listed in Tables S1 and S2; Table S3 defines the terminology, and Table S4 gives the AP of every RetinaNet arm.

Table S1: RetinaNet arm registry. "Calibration" lists the input distribution seen by entropy calibration; "sites" lists the quantized region.

Arm	Precision	Quantized sites	Calibration input
FP32	FP32	none	—
FP8-matched	FP8	full graph	clean, matched preprocessing
INT8-legacy	INT8	full graph	mismatched preprocessing (diagnostic)
INT8-matched	INT8	full graph	clean, 512 images
INT8-q95calib	INT8	full graph	J95 re-encodes, 512 images
INT8-selective	INT8	all except `/head/regression_head/`	clean, 512 images
INT8-sel-q95calib	INT8	selective	J95 re-encodes, 512 images
INT8-corruptcalib	INT8	full graph	fold-1 corruptions (noise, JPEG), 512 images
INT8-sel-corruptcalib	INT8	selective	in-family corruptions, 512 images
INT8-cc2calib	INT8	full graph	fold-2 corruptions (fog, motion blur), 512 images
INT8-sel-cc2calib	INT8	selective	fold-2 corruptions, 512 images
INT8-wonly	INT8	full graph; head *acts.* fp16	clean, 512 images
INT8-aonly	INT8	full graph; head *weights* fp16	clean, 512 images
FP8-legacy	FP8	full graph	pre-study build (context only)

The W-only and A-only arms are post-hoc QDQ-graph rewires of the frozen INT8-matched graph (Stage 2, Section 3.3 of the main text). The restored floating-point convolution operands and bypassed head weight initializers use fp16; engine I/O and some normalization intermediates retain fp32.

Table S2: FCOS-R50-FPN replication arm registry. The selective exclusion regex `^/head/regression_head/` covers the regression tower and the `bbox_ctrness` output, so the selective arm also dequantizes centerness. Because the score is $\sqrt{\sigma(\text{cls}) \cdot \sigma(\text{ctr})}$, the intervention is not treatment-identical to the RetinaNet selective arm, which touches regression outputs only. Calibration identical to the RetinaNet matched recipe.

Arm	Precision	Quantized sites	Calibration input
FP32	FP32	none	—
FP8-matched	FP8	full graph	clean, matched preprocessing
INT8-matched	INT8	full graph	clean, 512 images
INT8-selective	INT8	all except regression tower + centerness	clean, 512 images

Table S3: Terminology used in the main text and this supplement.

Term	Meaning
treatment / arm	complete recorded engine configuration / one treatment within an experiment
contract-matched	calibration preprocessing identical to inference
codec-matched	sharing the final JPEG quality-95 encoding
paired	evaluated on identical images and bootstrap resamples
cell	one corruption×severity condition (12 per evaluation)
J95 control	clean image re-encoded at JPEG quality 95
level deficit	AP shortfall an arm already carries on clean inputs
ΔE	change in an AP gap under corruption relative to the clean control
severity wall	a saturated cell where all arms sit near the accuracy floor
ledger	hash-addressed JSON record of inputs, predictions, or metrics
frozen / post hoc	fixed at the relevant protocol freeze / added after it
in-family / held-out	corruption families inside / outside a calibration fold

All INT8/FP8 engines are built by TensorRT 11.1.0.106 from ONNX graphs quantized by NVIDIA ModelOpt with entropy calibration (512-image budget; the study-built registries record version 0.45.0 and the pre-study YOLO11 ONNX registries predate that field; the diagnostic sweeps vary the estimator and include arms marked "128-img" in Table 2 of the main text, which use the earlier 128-image subset) with a 4,096 MiB builder workspace. Builder policy is recorded per engine and differs by program: the YOLO11 engines were built by `trtexec` before the frozen contract and carry TensorRT 11.1 defaults (TF32 enabled); the RetinaNet quantization arms built for this study use the TensorRT Python builder with TF32 disabled, while the original RetinaNet FP32 reference and

Table S4: RetinaNet arms under the frozen protocol (COCO-style AP on converted annotations). Corrupt: equal weight over the 12 corruption–severity cells. J95: codec-control clean; the FP32 and legacy arms predate the codec control (dashes). INT8-legacy also differs from the matched arms in builder policy (TF32 enabled by default versus disabled); Table 2 of the main text gives the controlled preprocessing-only comparison. Bold: the standard INT8 recipe and its selective repair.

	KITTI			VOC		
Arm	Clean	J95	Corrupt	Clean	J95	Corrupt
FP32 reference	50.15	–	37.80	57.23	–	42.32
FP8-matched	49.48	49.25	36.75	56.17	56.21	41.48
INT8-legacy (contract mismatch)	7.21	–	7.43	33.61	–	24.32
INT8-matched	**28.29**	**28.18**	**20.37**	**53.30**	**53.18**	**38.71**
INT8-selective (reg-head fp16)	**49.60**	**49.53**	**36.58**	**55.87**	**55.79**	**40.52**
INT8-q95calib (codec control)	28.26	27.85	20.42	53.33	53.39	38.54
INT8-sel-q95calib	50.28	49.79	36.94	55.81	55.95	40.40
INT8-corruptcalib	29.69	29.74	21.71	53.78	53.91	39.35
INT8-sel-corruptcalib	49.55	49.03	36.08	56.36	56.43	41.13
INT8-cc2calib (fold 2)	25.36	25.47	18.68	54.15	54.22	39.97
INT8-sel-cc2calib (fold 2)	49.72	50.54	37.20	56.40	56.43	41.58

the FP8/INT8-legacy arms are pre-study `trtexec` builds with TF32 enabled (all six retrained-holdout arms of Section S15 use the Python builder with TF32 disabled) (the ΔE contrasts only join quantized arms built under the same policy); the four FCOS arms, FP32 included, are TensorRT Python builder builds with TF32 disabled, so the FCOS block is internally policy-matched but not builder-identical to the pre-study YOLO11 engines. Calibration and evaluation images are disjoint. Checkpoints are one per dataset×model cell, and their selection provenance is disclosed: the KITTI and VOC YOLO11n/m/x models were fine-tuned from Ultralytics pretrained weights under the pilot program (`*_train_v1` runs, dataset YAMLs: KITTI `val: images/val`, VOC `val: images/test2007`), with the best checkpoint taken on the same partition that serves as the evaluation universe; COCO YOLO11 uses the official pretrained checkpoint without a dataset fine-tune; the torchvision RetinaNet-R50-FPN-v2 model is fine-tuned per dataset for 100 epochs at 640-pixel input (learning rate 10^{-4}, weight decay 10^{-4}, deterministic seed 20260807), selecting the best checkpoint by minimum validation loss on the same evaluation partition. FCOS-R50-FPN is trained under the same profile and checkpoint rule. All four RetinaNet/FCOS selected checkpoints are early picks: run-log epochs 21 (KITTI) and 5 (VOC) for RetinaNet, 14 (KITTI) and 10 (VOC) for FCOS (epoch counts here are the run log's 1-indexed values; the checkpoints store the same point 0-indexed). Training inputs are retained for reproduction: YOLO11 uses `configs/training/voc_kitti_yolo11_nmx_v1.json`, the driver `src/train_yolo_dataset.py` (queue `src/run_training_queue.py`), the dataset YAMLs in `configs/datasets/`, and the resolved Ultralytics arguments in `manifests/training/ultralytics_args/`; RetinaNet and FCOS use `src/train_retinanet_dataset.py` and `src/train_fcos_dataset.py` with `configs/training/cross_family_retinanet_r50_fpn_v2_v1.json`. All file hashes match the training manifests. The KITTI/VOC YOLO11 models instead select late best-fitness checkpoints (run epochs 65–100; recorded in `phase_a_arms/training/yolo11_best_fitness_epochs.json`) under Ultralytics' own rule, and the COCO YOLO11 models are the official pretrained weights with no study-side selection. Both FCOS runs completed all 100 scheduled epochs. On VOC the FCOS training loss diverged to NaN at epoch 70 (validation loss stayed finite through epoch 92, then also diverged); the epoch-10 selection point predates the divergence by 60 epochs, and no intervening epoch beat its validation minimum, so the divergence did not affect which checkpoint was exported. The training manifest records the schedule and selection rule but not the loss history; the per-epoch training/validation loss curves are shipped in the support package (`phase_b_results/training/` for RetinaNet, `phase_h_fcos_results/training/` for FCOS), and the selected epochs match the checkpoints' stored epoch fields (raw run logs are not redistributed; each curve record carries its source log's SHA-256). Per-run training manifests are hash-bound; the evaluation-partition overlap is a disclosed selection caveat (see the main text's limitations). The frozen condition schedule is 14 conditions per arm; the original RetinaNet FP32 and legacy arms predate codec control and carry 13.

S5 Inference and evaluation contract

Preprocessing differs by family. YOLO11 uses OpenCV bilinear resizing to a 640-pixel letterbox with centered padding of value 114 and RGB values divided by 255. RetinaNet and FCOS divide RGB values by 255, normalize with ImageNet means (0.485, 0.456, 0.406) and standard deviations (0.229, 0.224, 0.225), resize bilinearly to a longest side of 640 pixels, and zero-pad the bottom and right in normalized coordinates. YOLO11 blocks: confidence threshold 0.001, class-wise NMS at IoU 0.7, at most 300 detections. RetinaNet blocks: confidence threshold 0.05, at most 1,000 pre-NMS candidates per feature level, batched NMS at IoU 0.5, at most 300 detections. FCOS blocks:

confidence threshold 0.05 on the centerness-weighted score, at most 1,000 pre-NMS candidates per feature level, batched NMS at IoU 0.5, at most 300 detections. These are the RetinaNet constants, reused post hoc and frozen for FCOS, not torchvision's FCOS defaults (score threshold 0.2, NMS IoU 0.6, 100 detections per image); the decoder is a reimplementation of torchvision's post-processing, validated detection-for-detection against torchvision's own pipeline on 32 seeded synthetic head-output draws (`phase_h_fcos_results/decoder_validation.json`: 9,600 detections, maximum box and score deviation exactly zero). The check saturates the 300-detection cap on every draw, uses a square 640-pixel input at scale 1.0, and covers the 8-class KITTI head only. It validates the post-processing arithmetic, not letterbox un-mapping, sparse-output behavior, or end-to-end engine parity. All evaluations use the COCO protocol via `pycocotools` with `maxDets=100`; VOC and KITTI annotations were converted once from Ultralytics YOLO labels to COCO format and frozen; VOC objects marked difficult are omitted rather than kept as ignore regions. Decoder constants are listed per family in the paragraph above (identical for RetinaNet and FCOS) and held fixed within each; cross-family comparisons are comparisons of recipe bundles, never of formats alone.

S6 Corruption-aware calibration schedule

The intervention calibration sets reuse the same 512 images as the matched clean-calibration manifests. Each image is deterministically assigned, via a SHA-256-keyed hash of its identifier, to exactly one of six in-family cells, Gaussian noise or JPEG at severities 1, 3, or 5 (approximately 77–102 images per cell). Assigned corruptions use the frozen transform parameters above, and the outputs are terminally encoded at JPEG quality 95, matching the evaluation input contract. In this first fold fog and motion blur never appear in calibration and form the held-out split. A complementary second fold (schedule `nn-corruptcalib-fold2-v1`, frozen before any of its runs) assigns the same images to fog or motion blur at severities 1, 3, or 5 (75–95 images per cell) and holds out Gaussian noise and JPEG. Its pre-specified rules are (i) a held-out change from INT8-matched whose interval excludes zero (either direction counts; the main text reads it through the directional fold-1 criterion) and (ii) a positive held-out ΔE with an interval excluding zero, against both INT8-matched and INT8-q95calib.

The codec-matched control manifests encode the same 512 images at JPEG quality 95 with no named corruption, isolating codec adaptation from corruption coverage.

Table S5: Calibration-baseline decision rules and their registration status. "dir.": read directionally, so a significant harm counts as failure. "null-seek": met only when the interval contains zero; this bounds the residual but does not prove equivalence. [a]The registered wording is two-sided (interval excluding 0 in either direction); the main text uses the directional reading. [b]Registered as a primary estimand, not a success criterion. The codec-matched (q95calib) controls are post hoc for fold 1 and pre-specified for fold 2.

Contrast	Tier	Fold 1	Fold 2
Held-out gain vs clean-calib. arm	full	frozen, dir.	frozen, dir.[a]
Held-out ΔE vs matched/q95	full	post hoc	frozen, dir.
Held-out gain vs clean-calib. arm	sel.	frozen estimand[b]	post hoc
Held-out residual vs FP8 incl. 0	sel.	frozen, null-seek	post hoc

Table S5 lists the rules. Fold 1 has two success criteria: (i) the full-tier arm improves the held-out corrupt mean over INT8-matched with an interval above zero, supporting the range-mismatch hypothesis; (ii) the held-out mean of the selective arm is compatible with FP8-matched, which would let calibration explain the VOC residual. Criterion (ii) seeks a null and has no equivalence margin. The hypothesis concerns corrupted-input accuracy and cannot explain the clean deficit. The fold-2 replication rule is worded two-sidedly; it is read through the directional fold-1 criterion it cites, so a significant held-out decrease counts as failure, and the literal reading would change only the KITTI verdict. Two analyses were added after the fold-1 freeze: a comparison of held-out gains with the remaining deficit to FP8-matched, and the corruption-specific component $\Delta E_{\text{held-out}}$, which removes the shift that also appears on clean inputs. Reading this arithmetic interaction as a corruption-specific benefit assumes that the clean effect carries over additively; scale or floor effects can also produce a nonzero value (Section S7). Codec adaptation can shift AP in either direction, so held-out gains are attributed to corruption coverage only if they also hold against the codec-matched anchor.

S7 Per-arm calibration contrasts

Table S6 gives the full per-arm contrast set underlying Section 5.3 of the main text, and Fig. S1 groups the same contrasts by calibration arm and evaluation partition.

The additive ΔE decomposition is itself scale-sensitive (equal proportional changes can produce a nonzero additive interaction), so a retention-scaled check is reported: each fold's clean-control change is scaled by the

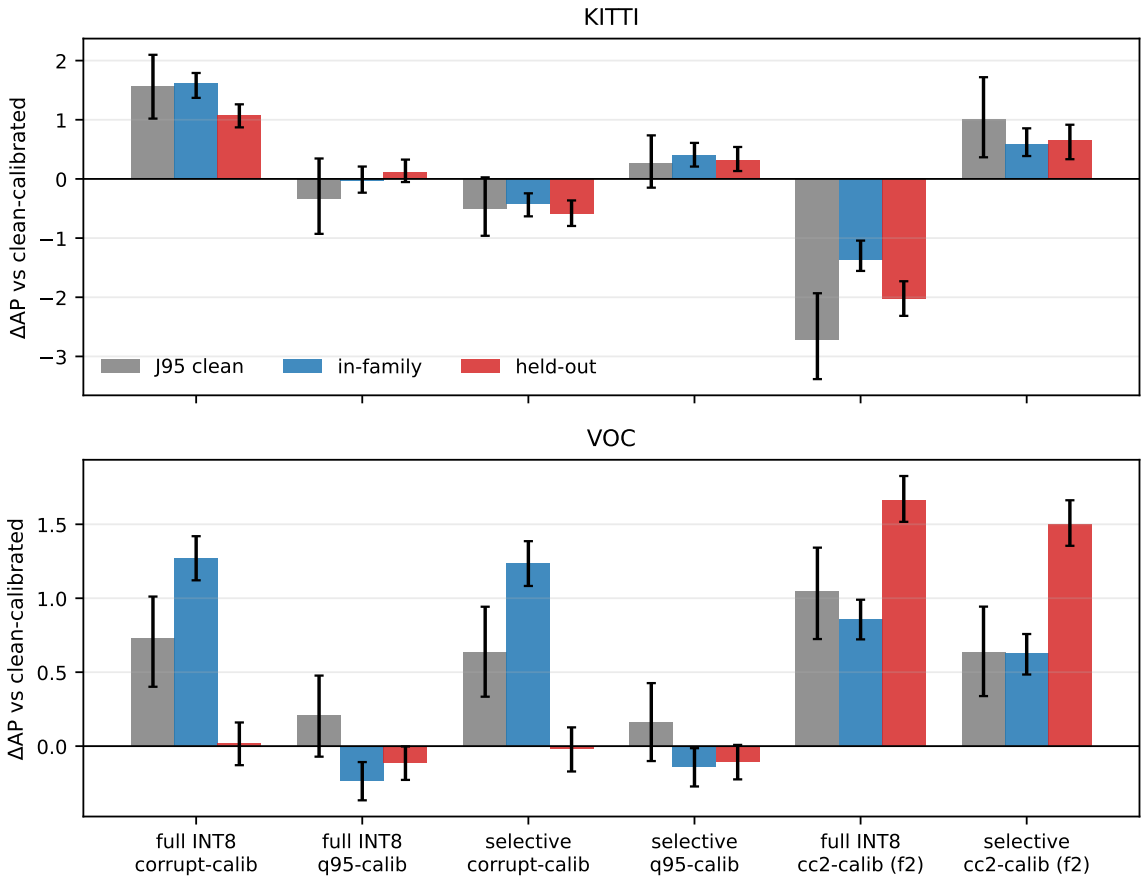


Figure S1: Calibration variants versus their clean-calibrated counterparts (AP points, 95% paired-bootstrap intervals). Grey: J95 clean control; blue: in-family corruptions; red: held-out corruptions; "f2" groups swap the family roles, and for the q95-calibration rows, which see no corrupted images, the in-family/held-out labels mark only the evaluation partition. Held-out bars that merely track the clean bar are not corruption-specific: that is the KITTI fold-1 pattern and, as a harm, the KITTI fold-2 full-INT8 pattern; the VOC fold-2 groups are the exception, where held-out clearly exceeds clean. For the q95-calibration arms, aggregate corrupted-AP changes stay below half an AP point; these changes reflect codec adaptation, not exposure to the named corruptions.

matched arm's AP retention on that fold's held-out families (0.63 for fold 1 and 0.82 for fold 2 on KITTI; 0.67 and 0.78 on VOC), predicting held-out changes of +0.98 and -2.22 on KITTI and +0.49 and +0.82 on VOC against the observed +1.07, -2.02, +0.02, and +1.66. Under this reading both KITTI cells are largely scale effects, VOC fold 1 still shows no corruption-specific benefit, and only VOC fold 2 exceeds its prediction materially (+1.66 vs. +0.82); anchoring on the codec arm gives the same qualitative answer (KITTI fold-1 residual -0.23 AP).

The fold-1 primary estimand also covers the selective-tier held-out gain over its clean-calibrated counterpart. This contrast (sel-corruptcalib minus sel) is -0.58 $[-0.80, -0.36]$ on KITTI (significantly negative, subject to the development-contamination caveat of the main text) and -0.01 $[-0.17, +0.13]$ on VOC. On KITTI fold 2 the selective arm is significantly lifted where the full-INT8 arm is harmed (J95 $+1.01$ $[+0.37, +1.72]$; held-out $+0.66$ $[+0.33, +0.92]$), but neither lift survives the codec anchor (sel-cc2calib minus sel-q95calib: J95 $+0.75$ $[-0.04, +1.56]$; held-out $+0.25$ $[-0.09, +0.52]$). Against FP8-matched, the VOC fold-2 selective arm's held-out residual ($+0.08$ $[-0.06, +0.23]$) is the only cell whose fold-1-style null-seeking interval includes zero; on KITTI the same contrast excludes zero favorably ($+1.04$ $[+0.64, +1.40]$). All fold-2 selective contrasts are post hoc.

S8 Paired bootstrap

Within each dataset×model block, one deterministic schedule of $B = 2{,}000$ resamples of image indices is shared across all arms and all conditions, so every contrast is evaluated inside common-image samples. Resamples index evaluated images by position, so a duplicated image contributes its detections and ground truth once per occurrence; this multiplicity is verified by a dedicated unit test (`tests/test_bootstrap_multiplicity.py`), which constructs a stub evaluation where resample $[0, 0, 1]$ scores 34/101 against the 51/101 that deduplication would give. Reported point estimates are full-sample plug-in values; reported intervals are the 2.5th–97.5th percentiles of

Table S6: RetinaNet calibration and codec-control contrasts (A minus B, AP points; plug-in point with 95% paired-bootstrap interval, 2,000 common-image resamples). J95 clean: codec-control clean cell; corrupt mean: 12 cells; in-family: Gaussian noise and JPEG (the fold-1 calibrated families); held-out: fog and motion blur (never calibrated in fold 1); for the q95-calibration rows, which see clean images only, these labels mark the evaluation partition rather than exposure; $\Delta E_{\text{held-out}}$ = held-out difference minus J95 clean difference, and $\Delta E_{\text{in-family}}$ is defined analogously; because every row here is an intervention-minus-baseline contrast (the ΔE definition of the main text), positive ΔE shifts the difference toward the first-named arm. For rows anchored to a clean-calibrated INT8 arm, a positive ΔE is a corruption-specific intervention benefit; for intervention arms compared with FP8-matched, substantial level deficits remain, and positive values describe gap narrowing under corruption, not a benefit. Fold-2 rows swap the family roles: their in-family is fog and motion blur and their held-out is Gaussian noise and JPEG. *: unadjusted 95% paired-bootstrap interval excludes zero.

Contrast (A − B)	Dataset	J95 clean	Corrupt mean	In-family	Held-out	$\Delta E_{\text{in-family}}$	$\Delta E_{\text{held-out}}$
INT8-corruptcalib − INT8-matched	KITTI	$+1.56^{*}$[+1.02, +2.10]	$+1.34^{*}$[+1.17, +1.48]	$+1.61^{*}$[+1.37, +1.79]	$+1.07^{*}$[+0.87, +1.26]	+0.05 [−0.54, +0.59]	−0.50 [−1.01, +0.03]
INT8-corruptcalib − INT8-matched	VOC	$+0.73^{*}$[+0.40, +1.01]	$+0.64^{*}$[+0.52, +0.76]	$+1.27^{*}$[+1.12, +1.42]	+0.02 [−0.13, +0.16]	$+0.54^{*}$[+0.24, +0.88]	-0.71^{*}[−1.02, −0.39]
INT8-q95calib − INT8-matched	KITTI	−0.34 [−0.93, +0.35]	+0.04 [−0.10, +0.22]	−0.02 [−0.23, +0.21]	+0.11 [−0.05, +0.33]	+0.31 [−0.35, +0.90]	+0.45 [−0.24, +1.04]
INT8-q95calib − INT8-matched	VOC	+0.20 [−0.07, +0.48]	-0.17^{*}[−0.26, −0.09]	-0.23^{*}[−0.37, −0.11]	-0.11^{*}[−0.228, −0.002]	-0.44^{*}[−0.73, −0.14]	-0.31^{*}[−0.62, −0.02]
INT8-corruptcalib − INT8-q95calib	KITTI	$+1.90^{*}$[+1.28, +2.54]	$+1.29^{*}$[+1.08, +1.45]	$+1.64^{*}$[+1.33, +1.86]	$+0.95^{*}$[+0.73, +1.15]	−0.26 [−0.89, +0.30]	-0.94^{*}[−1.60, −0.33]
INT8-corruptcalib − INT8-q95calib	VOC	$+0.52^{*}$[+0.22, +0.80]	$+0.81^{*}$[+0.70, +0.93]	$+1.50^{*}$[+1.35, +1.66]	+0.13 [−0.02, +0.28]	$+0.98^{*}$[+0.70, +1.31]	-0.39^{*}[−0.69, −0.07]
INT8-sel-corruptcalib − INT8-selective	KITTI	−0.50 [−0.96, +0.03]	-0.50^{*}[−0.65, −0.36]	-0.41^{*}[−0.63, −0.24]	-0.58^{*}[−0.80, −0.36]	+0.09 [−0.43, +0.52]	−0.08 [−0.68, +0.41]
INT8-sel-corruptcalib − INT8-selective	VOC	$+0.63^{*}$[+0.33, +0.94]	$+0.61^{*}$[+0.49, +0.72]	$+1.23^{*}$[+1.08, +1.39]	−0.01 [−0.17, +0.13]	$+0.60^{*}$[+0.29, +0.92]	-0.65^{*}[−0.97, −0.35]
INT8-sel-q95calib − INT8-selective	KITTI	+0.26 [−0.15, +0.74]	$+0.36^{*}$[+0.23, +0.49]	$+0.41^{*}$[+0.21, +0.61]	$+0.31^{*}$[+0.13, +0.54]	+0.15 [−0.39, +0.58]	+0.05 [−0.48, +0.57]
INT8-sel-q95calib − INT8-selective	VOC	+0.16 [−0.10, +0.43]	-0.12^{*}[−0.21, −0.04]	-0.14^{*}[−0.27, −0.01]	−0.11 [−0.22, +0.01]	-0.30^{*}[−0.60, −0.02]	−0.27 [−0.56, +0.02]
INT8-sel-corruptcalib − INT8-sel-q95calib	KITTI	-0.76^{*}[−1.35, −0.17]	-0.85^{*}[−1.05, −0.69]	-0.82^{*}[−1.08, −0.59]	-0.89^{*}[−1.16, −0.65]	−0.06 [−0.65, +0.54]	−0.13 [−0.83, +0.51]
INT8-sel-corruptcalib − INT8-sel-q95calib	VOC	$+0.47^{*}$[+0.16, +0.76]	$+0.73^{*}$[+0.61, +0.85]	$+1.37^{*}$[+1.22, +1.54]	+0.09 [−0.07, +0.25]	$+0.90^{*}$[+0.61, +1.22]	-0.38^{*}[−0.68, −0.08]
INT8-sel-corruptcalib − FP8-matched	KITTI	−0.23 [−1.12, +0.86]	-0.67^{*}[−0.91, −0.44]	−0.03 [−0.37, +0.31]	-1.30^{*}[−1.61, −1.00]	+0.20 [−0.74, +0.92]	-1.08^{*}[−2.18, −0.16]
INT8-sel-corruptcalib − FP8-matched	VOC	+0.21 [−0.06, +0.49]	-0.35^{*}[−0.46, −0.24]	-0.19^{*}[−0.32, −0.05]	-0.51^{*}[−0.66, −0.38]	-0.40^{*}[−0.67, −0.12]	-0.72^{*}[−1.00, −0.44]
INT8-corruptcalib − FP8-matched	KITTI	-19.51^{*}[−20.90, −18.04]	-15.04^{*}[−15.96, −14.17]	-16.74^{*}[−17.85, −15.66]	-13.34^{*}[−14.13, −12.57]	$+2.77^{*}$[+1.98, +3.48]	$+6.17^{*}$[+5.12, +7.10]
INT8-corruptcalib − FP8-matched	VOC	-2.31^{*}[−2.68, −1.98]	-2.13^{*}[−2.30, −1.97]	-2.13^{*}[−2.33, −1.93]	-2.12^{*}[−2.32, −1.96]	+0.17 [−0.10, +0.49]	+0.18 [−0.12, +0.50]
INT8-cc2calib − INT8-matched (fold 2)	KITTI	-2.71^{*}[−3.38, −1.93]	-1.69^{*}[−1.88, −1.44]	-1.35^{*}[−1.56, −1.04]	-2.02^{*}[−2.31, −1.73]	$+1.36^{*}$[+0.61, +2.16]	+0.69 [−0.10, +1.37]
INT8-cc2calib − INT8-matched (fold 2)	VOC	$+1.04^{*}$[+0.72, +1.34]	$+1.26^{*}$[+1.15, +1.37]	$+0.86^{*}$[+0.72, +0.99]	$+1.66^{*}$[+1.52, +1.83]	−0.19 [−0.49, +0.13]	$+0.62^{*}$[+0.30, +0.96]
INT8-cc2calib − INT8-q95calib (fold 2)	KITTI	-2.37^{*}[−2.90, −1.74]	-1.73^{*}[−1.96, −1.47]	-1.47^{*}[−1.68, −1.18]	-2.00^{*}[−2.32, −1.69]	$+0.91^{*}$[+0.28, +1.47]	+0.37 [−0.28, +0.90]
INT8-cc2calib − INT8-q95calib (fold 2)	VOC	$+0.84^{*}$[+0.54, +1.12]	$+1.43^{*}$[+1.32, +1.55]	$+0.97^{*}$[+0.82, +1.12]	$+1.89^{*}$[+1.75, +2.06]	+0.13 [−0.15, +0.43]	$+1.06^{*}$[+0.76, +1.38]
INT8-sel-cc2calib − INT8-selective (fold 2)	KITTI	$+1.01^{*}$[+0.37, +1.72]	$+0.62^{*}$[+0.42, +0.83]	$+0.59^{*}$[+0.39, +0.86]	$+0.66^{*}$[+0.33, +0.92]	−0.42 [−1.08, +0.24]	−0.35 [−1.09, +0.28]
INT8-sel-cc2calib − INT8-selective (fold 2)	VOC	$+0.63^{*}$[+0.34, +0.94]	$+1.06^{*}$[+0.96, +1.18]	$+0.63^{*}$[+0.48, +0.76]	$+1.50^{*}$[+1.35, +1.66]	−0.01 [−0.33, +0.29]	$+0.86^{*}$[+0.55, +1.18]
INT8-sel-cc2calib − INT8-sel-q95calib (fold 2)	KITTI	+0.75 [−0.04, +1.56]	$+0.27^{*}$[+0.08, +0.43]	$+0.28^{*}$[+0.03, +0.52]	+0.25 [−0.09, +0.52]	−0.47 [−1.25, +0.33]	−0.50 [−1.27, +0.33]
INT8-sel-cc2calib − INT8-sel-q95calib (fold 2)	VOC	$+0.48^{*}$[+0.18, +0.75]	$+1.19^{*}$[+1.07, +1.30]	$+0.74^{*}$[+0.59, +0.87]	$+1.64^{*}$[+1.49, +1.80]	+0.26 [−0.03, +0.56]	$+1.16^{*}$[+0.87, +1.49]
INT8-sel-cc2calib − FP8-matched (fold 2)	KITTI	$+1.29^{*}$[+0.64, +2.03]	$+0.45^{*}$[+0.20, +0.71]	−0.13 [−0.44, +0.20]	$+1.04^{*}$[+0.64, +1.40]	-1.42^{*}[−2.14, −0.76]	−0.25 [−0.93, +0.36]
INT8-sel-cc2calib − FP8-matched (fold 2)	VOC	+0.22 [−0.06, +0.48]	+0.11 [−0.01, +0.22]	+0.13 [−0.02, +0.26]	+0.08 [−0.06, +0.23]	−0.08 [−0.36, +0.20]	−0.14 [−0.40, +0.15]
INT8-cc2calib − FP8-matched (fold 2)	KITTI	-23.78^{*}[−25.30, −22.13]	-18.07^{*}[−19.07, −17.00]	-15.76^{*}[−16.63, −14.86]	-20.37^{*}[−21.58, −19.14]	$+8.02^{*}$[+7.05, +8.94]	$+3.41^{*}$[+2.51, +4.23]
INT8-cc2calib − FP8-matched (fold 2)	VOC	-1.99^{*}[−2.34, −1.66]	-1.51^{*}[−1.68, −1.35]	-1.28^{*}[−1.47, −1.12]	-1.74^{*}[−1.94, −1.53]	$+0.70^{*}$[+0.41, +1.01]	+0.25 [−0.04, +0.57]

the bootstrap distribution of each contrast. Intervals cover image resampling only; they do not cover training-seed, calibration-draw, engine-build, or corruption-realization variance, and image-level resampling treats frames as independent (KITTI frames from shared driving sequences are correlated, so those intervals may be optimistic).

Two-sided bootstrap p-values use a consistent plus-one convention in each tail, $\widetilde{p} = \min\{1,\, 2(b_{\min}+1)/(B+1)\}$, so the smallest attainable value at $B = 2{,}000$ is $2/2001 \approx 0.001$ and no p is ever exactly zero; Holm adjustment covers the nine paired YOLO11 blocks. Heterogeneity of ΔE across those blocks uses Cochran's $Q = \sum_i w_i(\hat{\Delta}_i - \bar{\Delta}_w)^2$ with $w_i = 1/\widehat{\mathrm{se}}_i^2$ from the bootstrap draws, $I^2 = \max\{0, (Q - \mathrm{df})/Q\}$, and the DerSimonian–Laird between-block standard deviation τ. Blocks within a dataset share images, so their estimates may be correlated; the cross-block covariance is not identifiable from the phase-B caches because each block's resample schedule is seeded per dataset×model. To bound the induced dependence, the nine YOLO11 blocks were rerun with $B = 1{,}000$ under a schedule seeded per dataset only (`shared_schedule/bootstrap/`): per-draw Pearson correlations of the FP8−INT8 ΔE between model scales are $r = \{0.240, 0.213, -0.104\}$ on KITTI (n∼m, n∼x, m∼x), $\{-0.033, -0.066, 0.111\}$ on VOC, and $\{0.057, 0.013, -0.196\}$ on the COCO subset. These are small and of mixed sign. Using this rerun's full within-dataset covariance matrix (blocks on different datasets share no images and are treated as independent; point estimates are the frozen plug-in values), the generalized Cochran's $Q = \mathbf{e}^\top \Sigma^{-1} \mathbf{e}$, with $\mathbf{e} = \hat{\boldsymbol{\Delta}} - \hat{\mu}_{\mathrm{GLS}}\mathbf{1}$ and $\hat{\mu}_{\mathrm{GLS}} = (\mathbf{1}^\top\Sigma^{-1}\hat{\boldsymbol{\Delta}})/(\mathbf{1}^\top\Sigma^{-1}\mathbf{1})$, is 67.1 on 8 df ($p \approx 2 \times 10^{-11}$), against 63.3 under independence; within-dataset values are 3.3 (KITTI, $p = 0.19$), 31.0 (VOC, $p \approx 2 \times 10^{-7}$), and 18.9 (COCO, $p \approx 8 \times 10^{-5}$) on 2 df, and the VOC YOLO11n−YOLO11m contrast has $z = 4.8$ with its covariance included (`analysis/nn_final_stats.py`, `gls_heterogeneity`). The dependence therefore does not change the heterogeneity conclusion, and it does not affect the per-block Holm adjustment, which requires valid marginal p-values but no independence among tests. RetinaNet and intervention intervals are per contrast.

As a cross-machine check, the pretrained-FCOS block was bootstrapped independently on two hosts under the same seed namespace (`coco__fcos_pretrained`): the 56 cell-level draw arrays agree to 2.2×10^{-16} absolute difference (identical seed, B, and image set), so resample schedules are portable; container file hashes still differ because NPZ serialization embeds metadata.

Point estimates are full-sample plug-in values computed from the metric records of each cell (for COCO, from the frozen 2,000-image subset evaluation); bootstrap draws supply intervals, standard errors, and p-values only. For the intervention, the corruption-specific component on the held-out family is $\Delta E_{\text{held-out}} = (\mathrm{AP}_A - \mathrm{AP}_B)_{\text{held-out}} - (\mathrm{AP}_A - \mathrm{AP}_B)_{\mathrm{J95}}$, the same paired construction as the main estimand. It is an arithmetic contrast on the AP scale: an equal additive intervention effect on clean and held-out inputs cancels exactly. Reading a nonzero value as a corruption-specific benefit requires the clean-level effect to transfer additively; scale and floor effects can otherwise produce a nonzero interaction, and the contrast does not identify a causal mechanism. The J95-clean and held-out levels are therefore reported alongside it.

S9 Sensitivity analyses

Table S7 resolves the paired ΔE of each YOLO11 block per corruption family (each entry is the family-mean FP8−INT8 gap minus the J95 gap) and gives the leave-one-out exclusion of the `motion_blur-s5` severity wall; Fig. S2 shows the per-family interactions. The sign heterogeneity is also *within* blocks: on VOC-x the JPEG family contributes −1.07 AP while Gaussian noise and motion blur contribute +0.69 and +0.67 AP; on COCO-m JPEG (+1.43) and motion blur (+0.99) contribute positively while noise contributes a null negative (−0.36). Motion blur is the most block-variable family of the four (across-block standard deviation ≈1.1 AP population, ≈1.2 sample), with JPEG close behind (≈1.0); neither is uniformly harmful or helpful. Excluding the severity wall leaves the cross-block heterogeneity essentially unchanged ($Q \approx 60.2$, $I^2 \approx 87\%$) and every Holm-surviving block's sign intact; the two KITTI blocks that do not survive Holm drift (YOLO11n's null point estimate flips from −0.01 to +0.15, and YOLO11m's unadjusted-significant negative turns null). The interaction pattern is therefore not an artifact of the saturated cell, although the marginal KITTI estimates are sensitive to it.

The same leave-one-out sensitivity applied to the intervention contrasts changes no frozen-criterion verdict; two secondary codec-control contrasts do move across the zero boundary on VOC (both on the held-out basis): INT8-corruptcalib minus INT8-q95calib goes from +0.13 [−0.02, +0.28] to +0.21 [+0.04, +0.38], and INT8-sel-q95calib minus INT8-selective goes from −0.11 [−0.22, +0.01] to −0.15 [−0.29, −0.01]. Both are read with the main text's no-multiplicity caveat.

The frozen configuration files are hash-identified: `nn_corruptcalib_v1_20260924.json` SHA-256 `f51011ca5d666b89...` and `nn_corruptcalib_fold2_v1_20260925.json` SHA-256 `5f2d122ae0a52d5f...` (full hashes in the archive manifest).

Freeze timing is established by local-file evidence rather than an external registry: the versioned configuration files' modification times and the `created_at_utc` fields in the calibration and quantized-ONNX manifests place the configurations before their arms' first inference (run manifests carry no timestamp field, and the git history records joint archival of configs and outcomes, not pre-inference ordering). The fold-1 file was written 2026-09-25 (its name retains the design date 20260924, and its `title` field says "frozen 2026-09-24"; the title records the design-freeze day, and the serialized file is one day later); the q95 codec-control schedules were added after the fold-1 results

were computed (the attempt identifier `nn_q95calib_v1_20260924` keeps the design-date convention and postdates the fold-1 runs in manifest time). The fold-2 primary rule is quoted verbatim: "`int8-cc2calib512 minus int8-matched512 on the fold-2 held-out mean (gaussian_noise+jpeg), paired bootstrap B=2000, same seed namespace as Phase B: CI95 excluding 0 => fold-1 frozen criterion replicated on fold 2`"; "replicated" is read through the directional fold-1 criterion it cites, so a significantly negative change counts as failure.

Table S7: Per-family ΔE (AP points; family-mean FP8−INT8 gap minus J95 gap) and the leave-one-out ΔE excluding `motion_blur-s5`, per dataset×model block. *: unadjusted 95% paired-bootstrap interval excludes zero.

Dataset	Model	Fog	Gaussian noise	JPEG	Motion blur	ΔE excl. mb-s5
KITTI	YOLO11n	$+1.11^{*}$	$+0.41$	$+0.17$	-1.71^{*}	$+0.15$ [-0.51, $+1.03$]
KITTI	YOLO11m	$+0.38$	-0.44	$+0.02$	-2.68^{*}	-0.35 [-0.98, $+0.29$]
KITTI	YOLO11x	-0.40	-1.98^{*}	-0.43	$+0.03$	-0.65^{*}[-1.08, -0.29]
VOC	YOLO11n	$+0.35^{*}$	$+0.28$	$+0.27$	$+0.34^{*}$	$+0.34^{*}$[$+0.09$, $+0.62$]
VOC	YOLO11m	-0.03	-0.48^{*}	-1.23^{*}	-0.56^{*}	-0.53^{*}[-0.76, -0.31]
VOC	YOLO11x	$+0.26^{*}$	$+0.69^{*}$	-1.07^{*}	$+0.67^{*}$	$+0.13$ [-0.06, $+0.35$]
COCO-2k	YOLO11n	$+0.17$	-0.22	-0.56^{*}	-0.69^{*}	-0.28 [-0.63, $+0.05$]
COCO-2k	YOLO11m	$+0.36^{*}$	-0.36	$+1.43^{*}$	$+0.99^{*}$	$+0.60^{*}$[$+0.27$, $+0.97$]
COCO-2k	YOLO11x	$+0.15$	-0.28	-2.10^{*}	$+0.13$	-0.59^{*}[-1.00, -0.26]

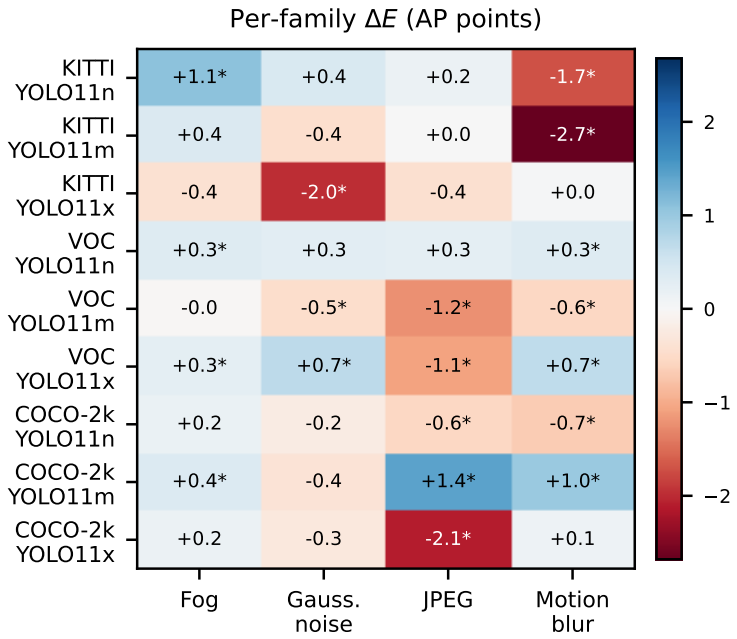


Figure S2: Per-family interaction ΔE (FP8 minus INT8, J95 basis) for the nine paired YOLO11 blocks, in AP. Star: unadjusted 95% bootstrap interval excludes zero. Every corruption family takes both signs across blocks. Two blocks have a single sign (VOC/YOLO11n positive, VOC/YOLO11m negative); the other seven mix signs, although the one positive cell of KITTI/YOLO11x is close to zero (+0.03).

Drive-clustered bootstrap for KITTI

KITTI object images are frames from a limited number of raw driving sequences, so image-level resampling can understate uncertainty. Every KITTI image was mapped to its raw drive with the official object devkit (`mapping/train_rand.txt` and `train_mapping.txt`; file hashes pinned in `src/build_kitti_drive_clusters.py`). The mapping is consistent with the Ultralytics file names: every image of a capture date has that date's camera resolution, and adjacent frames of the same drive have a median 16×16 average-hash distance of 18 bits, against 106 for random pairs. The 1,496-image evaluation set contains 108 drives. Every KITTI block was then rerun with a bootstrap that resamples whole drives (`N_BOOT=1000 bash src/run_nn_cluster_bootstrap.sh`; one shared schedule per block so that all arm contrasts stay paired; point estimates are unchanged). Table S8 compares the two intervals for the 34 KITTI contrasts reported in the main text and this supplement. Interval widths change by

a factor of 0.7–3.1 (median 1.4); drive resampling narrows some YOLO11 intervals and widens most RetinaNet and FCOS ones. Only 3 of 34 contrasts change significance (marked †), all with an image-bootstrap bound near zero: KITTI/YOLO11m ΔE, the FCOS FP8−INT8 clean contrast, and the FCOS FP32-minus-selective corrupt mean. The Holm-significant KITTI/YOLO11x block, the RetinaNet deficit and selective repair, the operand factorial, and the intervention verdicts are unchanged.

Table S8: KITTI interval claims under image-level ($B = 2{,}000$) and drive-clustered ($B = 1{,}000$) paired bootstrap, AP points. Width ratio: drive-interval width over image-interval width. †: the two intervals disagree on whether zero is excluded; *: the unadjusted 95% interval excludes zero. Fold-2 "held-out" families are Gaussian noise and JPEG.

Claim (KITTI)	Image bootstrap	Drive bootstrap	Width ratio
YOLO11n ΔE (FP8-INT8)	−0.01 [−0.67, +0.91]	−0.01 [−0.40, +0.66]	0.67
YOLO11m ΔE (FP8-INT8) †	−0.68*[−1.32, −0.05]	−0.68 [−0.90, +0.15]	0.83
YOLO11x ΔE (FP8-INT8)	−0.69*[−1.13, −0.33]	−0.69*[−1.71, −0.33]	1.72
RetinaNet FP8-INT8 matched, J95 gap	+21.07*[+19.57, +22.61]	+21.07*[+16.28, +24.54]	2.71
RetinaNet FP8-INT8 matched, ΔE	−4.69*[−5.49, −3.86]	−4.69*[−5.82, −3.36]	1.51
RetinaNet selective-matched, J95	+21.35*[+19.95, +22.81]	+21.35*[+17.12, +24.57]	2.61
RetinaNet FP8-selective, J95	−0.28 [−1.11, +0.56]	−0.28 [−1.60, +0.43]	1.21
RetinaNet FP8-selective, corrupt mean	+0.17 [−0.07, +0.40]	+0.17 [−0.29, +0.45]	1.58
RetinaNet FP8-selective, motion-blur ΔE	+1.30*[+0.51, +2.04]	+1.30*[+0.62, +2.32]	1.11
RetinaNet W-only-matched, J95	+21.25*[+19.88, +22.69]	+21.25*[+17.08, +24.50]	2.64
RetinaNet A-only-matched, J95	−0.04 [−0.20, +0.11]	−0.04 [−0.23, +0.12]	1.13
RetinaNet selective-W-only, J95	+0.09*[+0.01, +0.17]	+0.09*[+0.004, +0.183]	1.16
RetinaNet operand interaction, corrupt mean	+0.07*[+0.01, +0.14]	+0.07*[+0.02, +0.16]	1.13
Fold 1 corrupt-calib - matched, held-out	+1.07*[+0.87, +1.26]	+1.07*[+0.62, +1.34]	1.83
Fold 1 corrupt-calib - matched, held-out ΔE	−0.50 [−1.01, +0.03]	−0.50 [−1.25, +0.12]	1.31
Fold 2 corrupt-calib - matched, held-out	−2.02*[−2.31, −1.73]	−2.02*[−2.71, −1.50]	2.07
Fold 2 corrupt-calib - matched, held-out ΔE	+0.69 [−0.10, +1.37]	+0.69 [−0.48, +1.12]	1.08
Fold 2 INT8-sel-cc2calib - INT8-selective, held-out	+0.66*[+0.33, +0.92]	+0.66*[+0.06, +0.91]	1.46
RetinaNet FP8-selective, JPEG ΔE	−0.11 [−0.81, +0.50]	−0.11 [−1.12, +0.76]	1.44
RetinaNet A-only-matched, ΔE	+0.03 [−0.14, +0.20]	+0.03 [−0.16, +0.22]	1.17
RetinaNet operand interaction, J95	+0.14 [−0.04, +0.30]	+0.14 [−0.02, +0.29]	0.92
INT8-q95calib - matched, J95	−0.34 [−0.93, +0.35]	−0.34 [−0.76, +0.60]	1.07
Fold 1 corrupt-calib - FP8, held-out	−13.34*[−14.13, −12.57]	−13.34*[−15.37, −10.58]	3.06
Fold 1 INT8-sel-corruptcalib - FP8, held-out	−1.30*[−1.61, −1.00]	−1.30*[−1.86, −0.82]	1.69
Fold 1 INT8-sel-corruptcalib - INT8-selective, held-out	−0.58*[−0.80, −0.36]	−0.58*[−0.92, −0.33]	1.37
Fold 2 corrupt-calib - q95calib, held-out ΔE	+0.37 [−0.28, +0.90]	+0.37 [−0.51, +0.76]	1.08
Fold 2 INT8-sel-cc2calib - INT8-sel-q95calib, held-out	+0.25 [−0.09, +0.52]	+0.25 [−0.25, +0.46]	1.15
Fold 2 INT8-sel-cc2calib - FP8, held-out	+1.04*[+0.64, +1.40]	+1.04*[+0.74, +1.38]	0.84
FCOS FP8-INT8 matched, J95 †	−0.47*[−0.990, −0.001]	−0.47 [−2.35, +0.12]	2.50
FCOS FP8-INT8 matched, corrupt mean	−0.50*[−0.66, −0.36]	−0.50*[−0.94, −0.31]	2.09
FCOS FP8-INT8 matched, ΔE	−0.04 [−0.44, +0.40]	−0.04 [−0.51, +1.46]	2.34
FCOS selective-matched, J95	0.00 [−0.15, +0.12]	0.00 [−0.24, +0.11]	1.27
FCOS FP32-selective, J95	+0.07 [−0.25, +0.42]	+0.07 [−1.09, +0.44]	2.30
FCOS FP32-selective, corrupt mean †	+0.12*[+0.01, +0.22]	+0.12 [−0.16, +0.20]	1.65

S10 Evidence layers

Table S9 summarizes the evidence layers and their inferential status.

Table S9: Evidence layers and their inferential status.

Layer	Content	Inference
Diagnostic	Recipe sweeps that located the fragile region (Stage 2)	Single runs
Frozen paired	Nine YOLO11 blocks; RetinaNet matched and selective arms	Bootstrap; Holm over YOLO11
Calibration baseline	Two calibration folds with codec controls	Bootstrap, unadjusted
Historical	Earlier 144-cell YOLO11 grid	Context only
Post-hoc arms	Operand factorial, FCOS, retrained KITTI holdout	Bootstrap, unadjusted
Pretrained	COCO-pretrained RetinaNet and FCOS	Bootstrap, unadjusted
Audits	Max calibration, activations, Q/DQ, latency, rebuilds	Point estimates

Historical exploratory evidence. An earlier 144-cell exploratory analysis of the same YOLO11 engines and estimand covered four datasets (including TT100K), the full COCO codec-control pool, and additional arms, without the Holm family. It found an overall interaction of −0.02 AP $[-0.24, +0.15]$, −0.55 $[-0.78, -0.29]$ on post-selection VOC/KITTI holdouts with independently selected checkpoints, and 56 point-estimate sign disagreements between the raw corrupted-input gap and ΔE, with 17 ΔE intervals entirely below zero. These results motivated the frozen analysis and are not pooled with it. Two exploratory transfer cases, RT-DETR-L and legacy RetinaNet, failed in

INT8 under the unmigrated recipe; a corrected recipe was not tested for RT-DETR-L, which is therefore excluded from the main analysis.

Diagnostic arms (the RetinaNet decomposition: head-only and all-except-region variants) are single-point recipe sweeps produced during development and carry no intervals. Frozen paired evidence is the 14-condition paired protocol; its freeze covers the estimands, condition schedule, and bootstrap schedule (the nine-block Holm adjustment lives in the analysis code, not the frozen configuration), while the YOLO11 engines and evaluation partitions predate it and are shared with the historical layer. Calibration-baseline evidence is the corrupt-calibration experiment together with its codec-matched controls: the two corruption-calibration folds were frozen prospectively before their engines were built, while the fold-1 codec-control arms are post-hoc. Historical exploratory results (described above) are reported only as context and are never pooled with the frozen paired estimates. Post-hoc extensions are the operand-level factorial inside the RetinaNet regression head (W-only and A-only rewires of the frozen matched graph), the FCOS-R50-FPN replication, and the retrained KITTI holdout (Section S15): all were designed and executed after the frozen paired results were known, are evaluated on the same condition schedule with the same paired bootstrap estimand, and are reported as post-hoc extensions rather than frozen evidence. The FCOS phase package ships its own provenance (per-engine build manifests, quantized-ONNX manifests, per-run training manifests, and the decoder-validation record above) under `phase_h_fcos_results/`. The pretrained-checkpoint replication (torchvision RetinaNet-R50-FPN-v2 and FCOS-R50-FPN on a paired 2,000-image COCO val2017 subset) is likewise post hoc: its package ships quantized-ONNX manifests, TensorRT build manifests, per-cell input/prediction/metric records, and bootstrap draw caches under `phase_i_coco_pretrained_results/`; calibration used the hash-bound 512-image COCO train2017 manifest `coco_train_clean_512_s20260807_p0_v1_pilotroot` (canonical SHA-256 `bcac5aaa8b29...`). A final layer covers deployment-side audits and post-hoc diagnostics outside the paired estimand: the max-calibration counterfactual pair (`phase_j_maxcalib_results/`), the head-activation captures (`nn_actstats/`), the fivefold engine-rebuild ledger (`nn_rebuild_variance_v1_20260930/`), and the idle-gated latency ledgers (`nn_latency_v1_20260930/`). These diagnostics are descriptive and have no bootstrap intervals; the rebuild audit comprises five builds per engine.

S11 Artifact and audit boundary

Generated tables, numerical macros, and results figures are produced from retained ledgers by `analysis/build_nn_tables.py`; the protocol schematic is produced by `analysis/build_nn_framework_fig.py`. Engine registries, calibration manifests (including the corrupt-calibration cell-assignment schedule), input manifests, prediction manifests, metric records, and paired-bootstrap draw caches are SHA-256 hash-bound and retained in the accompanying compact-evidence package, deposited in the versioned research archive as release v3.0.5 (version DOI `https://doi.org/10.5281/zenodo.23331336`) under the concept DOI `https://doi.org/10.5281/zenodo.22031663`. Raw benchmark datasets, trained checkpoints, built TensorRT engines, and raw predictions are not redistributed; their provenance and regeneration path is documented in the archive. One binding asymmetry is disclosed for transparency: corrupted, codec-control, and YOLO11 original-clean cells bind each run to a content-addressed input manifest covering the encoded bytes (the YOLO11 clean runs bind the per-image clean manifests such as `kitti_val_clean_ultralytics_v1`), whereas the RetinaNet original-clean cells bind the frozen dataset split through its annotation SHA-256 (`clean-root:<hash>`). The matching per-image clean manifests exist in the archive and establish the reference bytes of each frozen split, but they are not bound into the RetinaNet clean run records; those runs therefore do not carry an execution-time byte-level binding for the clean inputs, only the annotation-level identifier.

S12 Deployment-side audits: latency and engine-build determinism

Two post-protocol audits measure deployment-side properties that the frozen accuracy contract does not cover. Both ran on the same RTX 5090 host and TensorRT 11.1.0.106 toolchain as the frozen engines and are retained with hash-bound provenance under `nn_latency_v1_20260930/` and `nn_rebuild_variance_v1_20260930/` in the evidence package. They are descriptive measurements on one host and enter no paired estimand.

Latency. Each of the 26 audited engines was timed in-process with CUDA events around `execute_async_v3` at batch 1 (200 warm-up plus 500 timed iterations per repetition, three repetitions; the harness enumerates GPU compute processes before every repetition and refuses to run while a non-allow-listed process holds the device). Table S10 reports per-engine medians and the across-repetition spread; for every engine, the median of the three within-repetition IQRs is at most 0.04 ms. Under this harness the INT8-matched engines run at 1.16–1.31 ms per image, the FP32 references that exist at 4.62–5.09 ms per image (a 3.9–4.0× speedup), and the selective-precision repair adds 1.3–6.4% over INT8-matched within the same detector–dataset pair. These are single-host medians: they bound the repair's deployment cost in the tested setting and do not transfer to other hardware, batch sizes, or power caps.

Table S10: Idle-gated batch-1 latency per engine (median of three repetitions, each the median of 500 timed iterations). Timing covers engine execution on a zero-filled input tensor and excludes preprocessing, data transfer, decoding, and NMS. Rep spread: maximum minus minimum of the three repetition medians.

Detector	Dataset	Arm	Median latency (ms)	Rep spread (ms)
FCOS (pretrained)	COCO	`fp32`	4.708	0.002
FCOS (pretrained)	COCO	`fp8-matched512`	1.368	0.024
FCOS (pretrained)	COCO	`int8-matched512`	1.179	0.017
FCOS (pretrained)	COCO	`int8-selective512`	1.224	0.007
RetinaNet (pretrained)	COCO	`fp32`	5.087	0.055
RetinaNet (pretrained)	COCO	`fp8-matched512`	1.494	0.005
RetinaNet (pretrained)	COCO	`int8-matched512`	1.307	0.012
RetinaNet (pretrained)	COCO	`int8-selective512`	1.325	0.016
FCOS	KITTI	`fp32`	4.615	0.054
FCOS	KITTI	`fp8-matched512`	1.356	0.033
FCOS	KITTI	`int8-matched512`	1.197	0.047
FCOS	KITTI	`int8-selective512`	1.219	0.017
RetinaNet	KITTI	`fp8-matched512`	1.359	0.056
RetinaNet	KITTI	`int8-aonly512`	1.188	0.012
RetinaNet	KITTI	`int8-matched512`	1.171	0.025
RetinaNet	KITTI	`int8-selective512`	1.225	0.033
RetinaNet	KITTI	`int8-wonly512`	1.188	0.017
FCOS	VOC	`fp32`	4.618	0.002
FCOS	VOC	`fp8-matched512`	1.365	0.035
FCOS	VOC	`int8-matched512`	1.162	0.011
FCOS	VOC	`int8-selective512`	1.236	0.036
RetinaNet	VOC	`fp8-matched512`	1.350	0.022
RetinaNet	VOC	`int8-aonly512`	1.215	0.003
RetinaNet	VOC	`int8-matched512`	1.192	0.040
RetinaNet	VOC	`int8-selective512`	1.247	0.037
RetinaNet	VOC	`int8-wonly512`	1.221	0.004

Engine-build determinism. Rebuilding each of seven engines five times from the same quantized ONNX under the frozen build policy (TF32 disabled, optimization level 3, 4096 MiB workspace) produced a distinct engine SHA-256 and a distinct inspector SHA-256 in all 35 builds, and the fused-layer count itself varies across rebuilds (counts span 270–309 layers across all audited builds; Table S11): under this builder the tactic autotuner does not settle on a bit-identical plan, or even the same fusion boundaries. The nondeterminism does not reach the metric in the evaluated slice. Across five rebuilds each of the KITTI INT8-matched and INT8-selective engines the measured AP is identical on both the clean and codec-control bases (spread 0.0000 AP). The evaluation coverage is two arms (matched and selective) on one dataset, so the audit bounds rather than eliminates build-to-build metric variance.

Table S11: Fivefold rebuild audit per arm: every rebuild is bit-distinct at both engine and inspector level. Layers: minimum–maximum fused-layer count. AP spread: maximum minus minimum AP across rebuilds, for original-clean and codec-control inputs; dashes mark unevaluated arms.

Arm	n	distinct eng. SHA	distinct insp. SHA	layers	AP spread
`coco_retinanet_int8-matched512`	5	5	5	285–309	–
`coco_retinanet_int8-selective512`	5	5	5	279–294	–
`kitti_retinanet_int8-aonly512`	5	5	5	270–274	–
`kitti_retinanet_int8-matched512`	5	5	5	279–298	0.0000; 0.0000
`kitti_retinanet_int8-selective512`	5	5	5	270–287	0.0000; 0.0000
`voc_retinanet_int8-matched512`	5	5	5	277–282	–
`voc_retinanet_int8-selective512`	5	5	5	277–286	–

S13 Head-region activation statistics

To probe the operand-factorial attribution at the distribution level, the activation tensors consumed by each RetinaNet and FCOS head were captured on the frozen 512-image clean-calibration manifests by running the unquantized ONNX graph in ONNX-Runtime (CUDA provider) with every head-consumed tensor exposed as a graph output. Per-channel $|x|_{\max}$ and excess kurtosis are computed over all spatial positions in the 512 captured images, with moments accumulated by streaming over every pixel. Each tensor's per-channel statistics are first summarized by their median, then the regional and per-level entries report medians across tensors. Each tensor is assigned to a region by graph consumption (tensors feeding both heads are "shared head inputs"). Captures cover the KITTI and VOC trained RetinaNet and FCOS graphs and the COCO pretrained RetinaNet graph (`nn_actstats/` in the evidence package; input bytes hash-verified against the calibration manifests).

Table S12 reports the summaries. On the three RetinaNet graphs the regression and classification heads have comparable median channel amplitudes, but regression-head activations have much heavier tails (on the KITTI graph, median excess kurtosis 10.6 versus 3.1 for the classification head, with per-channel maxima reaching 204795 versus 21826; on VOC, 8.0 versus 2.5; on the COCO pretrained graph, 6.0 versus 2.8). The tail is broad rather than a single stray channel: the median of the per-tensor channel-p95 excess kurtosis is 43.6 versus 9.1 on KITTI and

52.9 versus 6.7 on VOC. Rare large activations inflate a single per-tensor scale and spend the INT8 code budget on outlier coverage, which is exactly the signature under which activation quantization, not weight quantization, dominates the deficit (the operand factorial of Section 3.3 of the main text). Two cautions bound the reading: these are activations of the fp32 ONNX graph on calibration inputs, not the deployed quantizer's scale values (the calibrator's chosen activation amaxes on the matched INT8 graphs are recorded alongside, e.g. a median of 25.3 versus 12.5 on KITTI); and the captures are a measurement of the calibration distribution, which is itself what the quantizer saw. Tails under corrupted inputs are not measured here.

Table S13 resolves the same captures per FPN level, and a structural audit of the matched QDQ graphs (`nn_actstats/qdq_structure_*.json`) records how the head's activation ranges are parametrized. Two structure-level facts matter. First, the regression head does not share one activation scale across pyramid levels: the matched graph carries 25 regression-head-consumed activation quantizers (the five FPN feature tensors at the head input, shared with the classification head, plus four tower tensors at each of the five FPN levels), and every one is per-tensor with a level-specific scale, so a shared head-wide activation range is not the mechanism here. Second, the tail burden is level-ordered: regression-head median excess kurtosis is highest at the fine levels that carry small-object predictions (427.5 at P3 on KITTI, falling to 2.8 at P7; the classification head shows a much flatter profile, 20.1 to 0.1), and the same ordering holds on VOC; on the pretrained COCO graph the peak is at P4 (77.2 versus 61.0 at P3). Per-tensor grids at P3/P4 therefore spend the most code budget covering outliers. This fits the matched arm's deficit, which is concentrated in AP_S and AP_{75} (Table 9 of the main text), but it is a consistency argument, not a causal test.

The FCOS captures provide a cross-architecture falsification check for the extreme-tail profile, not evidence that FCOS activations are tail-free. Regression-head median excess kurtosis at P3/P4 is 14.4/11.3 on KITTI and 3.8/4.7 on VOC, compared with 427.5/208.1 and 179.1/143.0 for RetinaNet. In particular, FCOS still has positive excess kurtosis at KITTI P3; its fine-level tails are substantially lighter, not absent. The combination of lighter fine-level tails and no comparable head-specific deficit is consistent with the proposed distribution-level account, but FCOS also differs in architecture, anchor density, centerness scoring, and selective-output coverage, so the comparison does not isolate tail shape causally.

Table S12: Per-region activation statistics over the 512-image clean calibration sets for trained RetinaNet and FCOS graphs and the pretrained COCO RetinaNet graph. $|x|_{\max}$ is the per-channel absolute maximum; excess kurtosis is Fisher. For each tensor, channel-level statistics are summarized by a median; table medians are then taken across tensors. "Shared head inputs" are tensors consumed by both heads (FPN boundary features).

Graph	Region	n tensors	med. $\lvert x\rvert_{\max}$	max $\lvert x\rvert$	med. excess kurt.	max ex. kurt.
KITTI RetinaNet (trained)	regression head	192	8.60	7491	10.63	204795
	classification head	192	10.32	6156	3.14	21826
	shared head inputs	5	48.26	524	16.54	354
VOC RetinaNet (trained)	regression head	192	9.34	2290	7.99	261078
	classification head	192	8.34	1116	2.52	62412
	shared head inputs	5	74.33	576	5.87	96
COCO RetinaNet (pretrained)	regression head	192	7.57	1282	6.04	313956
	classification head	192	7.52	815	2.79	21296
	shared head inputs	5	50.92	270	3.06	43
KITTI FCOS (trained)	regression head	217	2.22	49	1.17	59256
	classification head	192	5.29	89	1.39	84352
	shared head inputs	5	1.18	5	10.29	61
VOC FCOS (trained)	regression head	217	2.45	183	1.44	373590
	classification head	192	7.61	200	1.81	3276795
	shared head inputs	5	9.37	54	5.82	26

Table S13: Per-FPN-level activation statistics, reported as the median of per-tensor median-channel $|x|_{\max}$ / excess kurtosis at each level (P3 finest, P7 coarsest; shared FPN boundary tensors are excluded from these head-specific summaries). The table includes three RetinaNet and two FCOS graphs; fine-level tails are substantially heavier in RetinaNet.

Graph	Head	P3	P4	P5	P6	P7
KITTI RetinaNet	reg.	22.6/427	14.6/208	10.5/81	6.7/8	3.6/3
	cls.	19.6/20	15.8/19	10.2/9	6.3/2	3.5/0
VOC RetinaNet	reg.	29.9/179	18.5/143	10.5/76	7.1/28	4.6/8
	cls.	11.6/4	9.9/5	10.2/6	7.3/3	4.7/1
COCO RetinaNet (pretrained)	reg.	21.3/61	14.1/77	9.2/41	6.1/20	4.1/8
	cls.	10.6/3	9.2/4	7.7/4	6.0/2	4.2/1
KITTI FCOS	reg.	5.2/14	3.4/11	2.3/6	1.0/0	0.6/-0
	cls.	13.3/11	10.9/9	7.5/3	3.2/0	2.2/-1
VOC FCOS	reg.	3.3/4	3.3/5	2.7/5	1.8/2	0.8/-0
	cls.	10.9/4	11.5/6	9.3/6	6.3/3	2.4/-1

S14 Max-calibration counterfactual

Section S13 shows the regression-head activations are heavy-tailed under the entropy calibrator's inputs, which admits two distinct cost accounts: tail clipping (the chosen range excludes outliers) or bin resolution (the range is fine but wide relative to the signed INT8 grid spacing). A max-calibration arm changes clipping and resolution jointly: it sets each activation range bound to the absolute maximum observed on the calibration set, removing calibration-set clipping at the price of the coarsest bin resolution (evaluation-time activations beyond that maximum can still clip). A post-hoc counterfactual pair was therefore run, INT8-matched and INT8-selective under the max estimator with the same graphs, frozen 512-image clean-calibration manifests, and 14-condition evaluation schedule, on the KITTI and VOC RetinaNet detectors (`nn_maxcalib_v1_20260930`; single engine builds, point estimates only, no bootstrap draws).

Table S14 reports the levels. Graph-wide max calibration repairs neither dataset. On KITTI, max calibration does not repair the matched deficit but deepens it, from 28.29 to 14.02 AP on the clean basis, while the head-exclusion benefit persists under the new estimator (selective minus matched 9.47 AP clean, 5.51 AP corrupt-mean), so the regression-head sensitivity is not an artifact of the entropy estimator. On VOC the max estimator is catastrophic on *both* arms (Table S14; clean AP ≈ 0.001–0.002): the detection-score distribution collapses (the 99th-percentile detection score on the clean run drops from ≈ 0.91 under entropy calibration to ≈ 0.15; `voc_score_audit.json` in the evidence package), consistent with a failure in the score (classification) pathway that regression-head exclusion cannot rescue. A 128-image draw of the same estimator in the earlier recipe sweep scored 23.94 AP on the KITTI clean basis. The two calibration budgets give different AP values; variability across repeated calibration draws at a fixed budget was not measured.

Because the graph-wide swap perturbs every quantizer, it cannot separate a head-local estimator effect from collateral damage elsewhere (the VOC collapse is the extreme case). A second counterfactual was therefore run, restricted to the regression head's inputs (`nn_maxreg_v1_20261001`; same single-build, point-estimate-only protocol): starting from the frozen INT8-matched graphs, the entropy-calibrated scales of every activation quantizer consumed by a regression-head node (25 per graph, per-tensor: 20 inside the tower, one per conv per FPN level, plus the five FPN-boundary quantizers at the head input; Section S13) were replaced by the calibration-set maxima measured on the same captured activations, while every other quantizer (backbone and classification tower) kept its entropy-calibrated scale byte-identical. The five boundary quantizers are shared with the classification head, so the patch also changes the classification head's input scales; it is regression-head-local inside the tower but not at the shared boundary. The patch is implemented by `src/patch_maxreg_scales.py`: each selected quantizer's Q and DQ scale becomes $\mathrm{fp16}(|x|_{\max}/127)$, with $|x|_{\max}$ the largest per-channel absolute maximum captured for that tensor (Section S13). Re-running the script on the frozen matched graphs reproduces the archived patched graphs exactly: byte-identical on KITTI, and identical nodes and initializers on VOC (the file hash differs only through serialization; `phase_k_maxreg_results/patch_reproduction.json`). This arm leaves VOC functional and still does not repair the deficit (Table S14): original-clean AP sits at 27.15 AP on KITTI and 52.32 AP on VOC, against 28.29/53.30 for the entropy-matched arms, and the corrupt-mean levels (19.58/37.93 AP) track the entropy-matched arms within about one AP point. Widening the regression-head-consumed ranges to the observed maxima recovers nothing.

The interpretation is bounded. Extending ranges to the observed calibration maxima, graph-wide or restricted to regression-head-consumed quantizers, does not improve AP and can destroy the classifier outright. Because max calibration reduces calibration-set clipping and coarsens the grid at the same time, these experiments do not separate a clipping benefit from the accompanying resolution cost; they are compatible with a resolution account (a fixed signed INT8 grid over a wide range) but do not establish it. Resolution is also not separated from the other numerical changes a coarser scale induces (larger rounding bias on in-range values, dequantization shift), and max calibration does not preclude evaluation-time clipping of activations beyond the calibration-set maximum. Corruption-aware calibration and max calibration together bracket the range-choice dimension: neither repairs the deficit, and the regression-head benefit survives both.

S15 KITTI holdout with a retrained checkpoint

The original KITTI checkpoints were trained on the full 5,985-image Ultralytics training partition, which contains the 1,197 images of the later resplit's final partition. A holdout was therefore built by retraining RetinaNet-R50-FPN-v2 with the frozen profile (100 epochs, seed 20260807) on the 4,788-image resplit training list, selecting the checkpoint with the lowest loss on the 1,496-image selection set (epoch 12), and rebuilding all six arms with the TensorRT Python builder (TF32 disabled), the five quantized arms using the frozen recipes and a 512-image calibration list drawn from the training list (`kitti_confirmatory_train_512_s20260818_v1`; no overlap with the final partition). All arms were evaluated on the final partition and, for reference, on the selection set under the 14-condition schedule (`src/run_nn_kitti_holdout_v2_queue.sh`). Intervals use the paired image bootstrap ($B = 2{,}000$) and the drive-clustered bootstrap ($B = 1{,}000$; `src/run_nn_kitti_holdout_v2_bootstrap.sh`). The final images share drives with the training list, so the holdout is image-disjoint but not sequence-disjoint. Table S15

Table S14: Max-calibration counterfactual on RetinaNet (post-hoc diagnostic, single builds, plug-in AP points without intervals). Reference rows repeat the frozen entropy-calibrated paired-protocol arms for the same detector–dataset pairs; bold rows are the max-estimator arms on the same 14-condition schedule: graph-wide matched/selective, then the variant with the max estimator applied only to activation quantizers consumed by the regression head (including the five FPN-boundary quantizers shared with the classification head). "Orig. clean" is the original clean image; "J95" is the JPEG quality-95 codec control.

Dataset	Arm	Orig. clean	J95 (codec ctrl.)	corrupt mean
KITTI	FP32 reference	50.15	–	37.80
	FP8-matched (entropy)	49.48	49.25	36.75
	INT8-matched (entropy)	28.29	28.18	20.37
	INT8-selective (entropy)	49.60	49.53	36.58
	INT8-matched (max)	14.017	14.176	7.817
	INT8-selective (max)	23.484	24.072	13.326
	INT8-matched (max, reg.-consumed quantizers)	27.148	27.429	19.584
VOC	FP32 reference	57.23	–	42.32
	FP8-matched (entropy)	56.17	56.21	41.48
	INT8-matched (entropy)	53.30	53.18	38.71
	INT8-selective (entropy)	55.87	55.79	40.52
	INT8-matched (max)	0.001	0.002	0.005
	INT8-selective (max)	0.002	0.002	0.005
	INT8-matched (max, reg.-consumed quantizers)	52.316	52.162	37.929

reports arm levels and contrasts.

Table S15: RetinaNet on the KITTI holdout (1,197 images unseen by the retrained checkpoint). Top: arm levels in AP on the final holdout and, for reference, on the selection set; "Orig. clean": original clean images; "Corrupt": equal-weight mean over the 12 corruption–severity cells. Bottom: paired contrasts with 95% intervals from the image bootstrap and, on the J95 basis, the drive-clustered bootstrap. *: unadjusted interval excludes zero.

	Final holdout			Selection set	
Arm	Orig. clean	J95	Corrupt	J95	Corrupt
FP32	42.19	42.20	32.26	44.50	34.02
FP8-matched	41.41	41.54	31.34	43.56	33.16
INT8-matched	27.44	27.49	21.05	29.01	22.16
INT8-selective	41.50	41.34	31.46	43.70	33.07
INT8 W-only	41.50	41.38	31.45	43.74	33.06
INT8 A-only	27.29	27.47	20.96	28.94	22.13

Contrast (final holdout)	J95, image	J95, drive	Corrupt mean, image
FP8-matched − INT8-matched	$+14.06^{*}$[+12.85, +15.35]	$+14.06^{*}$[+10.92, +16.60]	$+10.29^{*}$[+9.49, +11.16]
INT8-selective − INT8-matched	$+13.85^{*}$[+12.76, +15.00]	$+13.85^{*}$[+10.98, +16.28]	$+10.41^{*}$[+9.63, +11.23]
FP8-matched − INT8-selective	+0.21 [−0.48, +1.11]	+0.21 [−0.47, +0.68]	−0.12 [−0.33, +0.10]
W-only − INT8-matched	$+13.90^{*}$[+12.80, +15.01]	$+13.90^{*}$[+11.03, +16.43]	$+10.40^{*}$[+9.62, +11.21]
A-only − INT8-matched	−0.02 [−0.22, +0.18]	−0.02 [−0.17, +0.15]	-0.09^{*}[−0.16, −0.04]
INT8-selective − W-only	−0.05 [−0.21, +0.11]	−0.05 [−0.20, +0.04]	+0.01 [−0.02, +0.05]
FP32 − INT8-selective	$+0.87^{*}$[+0.21, +1.59]	$+0.87^{*}$[+0.27, +1.24]	$+0.81^{*}$[+0.59, +1.00]

An earlier post-hoc check (`nn_holdout_v1_20261001`) evaluated the original checkpoint's four factorial arms on the same 1,197 images. Those images were part of that checkpoint's training data, and 95 of them were in its calibration list, so the check is a seen-image consistency check, not a holdout; its absolute AP is correspondingly much higher (Table S16).

S16 Minimum reporting checklist for paired corruption studies

- record the complete treatment recipe (checkpoint, quantizer coverage, calibration list and preprocessing, builder settings, decoder), not just the datatype label;
- pair treatments on identical image bytes, including terminal encoding;
- subtract the matched clean gap before reading a corruption interaction;
- report absolute corrupted AP and the clean gap alongside every interaction, so floor effects cannot masquerade as robustness;
- keep calibration corruption families disjoint from held-out evaluation families when claiming calibration-based robustness;
- carry codec-matched calibration controls before attributing calibration gains to corruption coverage;
- flag severity-wall cells where every treatment including FP32 is at the floor, and report a leave-one-out sensitivity;
- audit engine-build determinism before treating a serialized engine as a fixed artifact; tactic autotuning makes rebuilds bit-distinct, so hash the engine and inspector and spot-check metric spread;

Table S16: Original KITTI checkpoint: factorial arms on the 1,197 final images (seen in training; 95 also in the calibration list) and on the selection set. Plug-in AP without intervals. "Orig. clean": original clean image; "J95": JPEG quality-95 codec control.

Split	Arm	Orig. clean	J95 (codec ctrl.)	corrupt mean
Final (seen)	INT8-matched	34.70	34.42	24.83
Final (seen)	INT8-selective	65.25	65.17	46.01
Final (seen)	INT8 W-only	65.30	64.90	45.99
Final (seen)	INT8 A-only	34.64	34.71	24.86
Selection	INT8-matched	28.29	28.18	20.37
Selection	INT8-selective	49.60	49.53	36.58
Selection	INT8 W-only	49.50	49.44	36.52
Selection	INT8 A-only	28.31	28.14	20.36

- gate latency claims on an idle-device harness with repetitions and dispersion, and scope them to the measured host.